

LANGUAGE: C++ PROGRAMMING LEVEL: INTERMEDIATE-ADVANCED

Day	Topic	Subtopics	Interview Coding Problems (LeetCode / HackerRank)	End of Module Use Case
Day 1	Revision of C++ Basics & OOP	C++ Program Structure, Data Types, Functions, Arrays, Strings, Classes & Objects, Constructors, Destructors, Encapsulation, Inheritance, Polymorphism Revision	LeetCode: Two Sum, Valid Parentheses, Maximum Subarray HackerRank: Classes and Objects	Student Result Analysis System
Day 2	Advanced Pointers & Memory Management	Pointers in C++, References, Smart Pointers (unique_ptr, shared_ptr), Pointer Arithmetic, Dynamic Memory Allocation, new/delete, Memory Leaks, RAII	LeetCode: Move Zeroes, Remove Element, Linked List Cycle HackerRank: Pointer Exercises	Dynamic Memory-Based Employee Record System
Day 3	Advanced OOP Concepts	Copy Constructor, Move Constructor, Deep Copy vs Shallow Copy, Operator Overloading, Friend Functions, Friend Classes, this Pointer, Static Members	LeetCode: Design HashMap, Merge Sorted Array HackerRank: Inherited Code	Banking Management System
Day 4	Templates & Generic Programming	Function Templates, Class Templates, Template Specialization, Variadic Templates, Generic Programming Concepts	LeetCode: Kth Largest Element, Binary Search HackerRank: C++ Templates	Generic Data Processing Framework
Day 5	STL Containers & Iterators	vector, deque, list, stack, queue, priority_queue, set, map, unordered_map, Iterators, Container Adapters	LeetCode: Top K Frequent Elements, Group Anagrams HackerRank: STL Vector, Maps-STL	Online Shopping Cart System
Day 6	STL Algorithms & Lambda Expressions	sort(), find(), count_if(), transform(), accumulate(),	LeetCode: Sort Colors, Merge Intervals	Student Ranking & Analytics System

		Lambda Functions, Custom Comparators, Predicate Functions	HackerRank: STL Algorithms	
Day 7	File Handling & Serialization	File Streams (ifstream, ofstream, fstream), Binary Files, Serialization, Deserialization, CSV & JSON Basics, Exception Handling in File Operations	LeetCode: Matrix Diagonal Sum, Valid Sudoku HackerRank: File I/O	Persistent Student Database System
Day 8	Exception Handling & Debugging	try-catch-throw, Custom Exceptions, Stack Unwinding, Assertions, Debugging Techniques, GDB Basics, Logging Concepts	LeetCode: Divide Two Integers, Single Number HackerRank: Exceptional Server	Secure Transaction Processing System
Day 9	Data Structures using C++	Linked List, Stack, Queue, Circular Queue, Trees, Binary Search Tree, Heap Basics	LeetCode: Invert Binary Tree, Min Stack HackerRank: Queue using Two Stacks	Library Management & Tracking System
Day 10	Advanced Algorithms & Problem Solving	Recursion Optimization, Backtracking, Greedy Algorithms, Sliding Window, Two Pointer Technique, Binary Search Optimization	LeetCode: Combination Sum, Subsets, Search in Rotated Sorted Array HackerRank: Problem Solving Medium	Route Optimization Engine
Day 11	Multithreading & Concurrency	Threads, Mutex, Synchronization, Deadlocks, Race Conditions, async, Futures, Thread Pools	LeetCode: Print in Order, FizzBuzz Multithreaded HackerRank: Threading Basics	Multi-threaded Log Processing System
Day 12	Socket Programming & System Concepts	TCP/IP Basics, Client-Server Architecture, Socket APIs, Network Communication, Process vs Thread, IPC Concepts	LeetCode: Number of Islands, Flood Fill HackerRank: Networking Basics	Client-Server Chat Application
Day 13	Advanced Interview Coding + RDBMS Concepts	Time Complexity, Space Complexity, Hashing, Optimization Techniques, Mock Coding Round, SQL Basics, Joins, Normalization, Aggregate Functions	LeetCode: Best Time to Buy & Sell Stock, Valid Anagram, Merge Intervals HackerRank: SQL Basic Select, Sparse Arrays	Student Database Management System

Day 14	Modern C++, Cloud & DevOps Integration	C++11/14/17 Features, Move Semantics, auto, constexpr, Build Systems (CMake), Git & GitHub, Docker, Kubernetes, CI/CD Pipelines	LeetCode: Product of Array Except Self, Single Number HackerRank: Linux Shell Basics, Git Tutorials	End-to-End Enterprise Deployment Pipeline using GitHub + Docker + Kubernetes
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